Dipannoy Das Gupta

dg47@email.sc.edu | +1 (575) 442-8905 | Columbia, SC

HIGHLIGHTS

AI researcher with 3+ years of experience building machine learning models that discover and design new materials, replacing slow and expensive lab experiments with fast, accurate predictions.

EDUCATION

PhD Candidate, Department of Computer Science, University of South Carolina — *2022 – Present*
M.Sc. in Computer Science (En-route), University of South Carolina — *2022-2025*
B.Sc. in Computer Science & Engineering, — *2015-2019*
Bangladesh University of Engineering & Technology (BUET)

SKILLS

- **Research:** Graph Neural Networks (GNN), Predictive Modeling, Diffusion Models, Large Language Models (LLM), Domain Adaptation, Inverse Design, Self-Supervised Learning (SSL), Atomistic Simulations (Pymatgen, ASE)
- **ML Libraries:** PyTorch, DGL, TensorFlow, Keras, Scikit-Learn, Weights & Biases (W&B)
- **Languages & Frameworks:** Python, C#, C++, JavaScript, Node.js, React, ASP.Net Core, SQL
- **Tools & Systems:** Git, Linux, High Performance Computing (HPC), Slurm, CUDA

RESEARCH EXPERIENCE

Graduate Research Assistant, University of South Carolina — *Aug 2022 – Present*

- **Multi-Property Inverse Design:** Built a framework that predicts several competing material properties at once, thermodynamic stability, band gap, and mechanical stiffness, instead of one at a time, and uses them to map out the best trade-off materials for a given design goal. Sharing one model across properties cut prediction error by up to 68% versus training a separate model per property, and the resulting trade-off maps matched true lab-measured results.
- **Crystal Structure Prediction:** Developed a generative AI model that predicts a stable crystal structure from a chemical formula in a single step, instead of the hundreds to thousands of steps competing methods need. It outperformed the leading prior methods on accuracy (83.6% vs. 78.3% and 77.9%) while running 20-200 times fewer computations, generating 10,000 structures in under 1.5 minutes versus over an hour for the next-best approach.
- **Cooperative GNN for Materials:** Designed a graph neural network that automatically learns which atomic bonds actually matter and prunes the rest, removing up to 38% of chemically redundant connections. This adaptive message-passing approach improved band gap prediction accuracy by up to 2.8% and matched or improved formation-energy accuracy using far fewer connections, while avoiding a common failure where deep models lose the ability to tell atoms apart.
- **Conditional Generative Models:** Examined diffusion models for inverse materials design using equivariant GNN decoders with different conditioning schemes. Implemented adapter-based conditioning, node-based conditioning, and skewed-dataset fine-tuning to generate valid crystals with target bandgaps showing up to 5 times improvement in control and novelty in generation. Designing materials with a target bandgap directly impacts the performance of LEDs and semiconductor devices.
- **Domain Adaptation for Materials:** Implemented parameter-based domain adaptation across GNN models - SchNet, PAiNN, and ALiGNN; improved model accuracy up to 1.5-2 times for screening new materials.

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer, Edusoft Consultants Ltd., Dhaka, Bangladesh — *Jun 2019 – Dec 2021*

- Designed database schemas and implemented backend APIs for academic scheduling, course management, student enrollment workflows, tuition billing and payment processing.
- Developed data ingestion and transformation pipelines for institutional records; performed data validation, normalization, and migration across legacy systems.
- Optimized backend queries and system workflows to improve response time and reduce scheduling conflicts in academic management systems.